GAS LEAKAGE DETECTOR

Project Report

Prepared By
Pranav
Class 6 - Section C
Gita Bharathi School of Excellence
Mysuru

Abstract

The Gas Leakage Detector is a science project that demonstrates how sensors can improve safety by detecting gases and vapors in the surrounding air. The system uses an MQ-2 gas sensor to continuously monitor environmental conditions. When the sensor reading exceeds a preset threshold, an LED indicator and buzzer are activated to provide an immediate warning. This project introduces students to sensors, electronics, and automation while highlighting a real-world safety application. It is easy to build, simple to demonstrate, and educational in nature. Although the MQ-2 can respond to multiple gases, it effectively demonstrates the principle of early gas detection and automated alert systems.

Detailed Project Description

Gas leakage can be a serious safety concern because some gases and vapours may be difficult to detect quickly, especially when people are not nearby. If not identified in time, gas leaks can create unsafe conditions and increase the risk of accidents. A Gas Leakage Detector is designed to demonstrate how technology can provide an early warning when changes in air quality are detected.

The project uses an **MQ-2 gas sensor** as its main component, along with an **LED indicator**, **buzzer**, and **power supply**. The MQ-2 is a general-purpose gas sensor that responds to different gases and vapours present in the environment. It continuously monitors the surrounding air and generates a sensor reading based on gas concentration levels. The system compares this reading with a preset threshold value. When the reading exceeds the defined limit, the LED indicator lights up and the buzzer sounds, providing both visual and audible alerts.

The primary objective of this project is to demonstrate how sensors and electronic systems can be used to address real-world safety challenges. The detector provides an immediate warning when elevated levels of gas or vapour are detected, helping to increase awareness of potentially unsafe conditions. The project also introduces important concepts related to sensor technology, electronic circuits, automation, and engineering design.

One of the key advantages of this project is its simplicity. It is cost-effective, easy to assemble, and suitable for classroom demonstrations and science exhibitions. However, the MQ-2 sensor is not LPG-specific and can respond to multiple gases and vapours. Therefore, a sensor response does not confirm the presence of LPG alone. For this reason, the project should be considered an educational demonstration model rather than a certified household safety device.

Overall, the Gas Leakage Detector demonstrates how science, electronics, and problem-solving can be combined to create a practical safety-focused system. It highlights the importance of early-warning technology and the role of sensors in improving awareness and promoting safer environments.